PP-C-401c
February 21, 1966

SUPERSEDING
Int. Fed. Spec. PP-C-00401b (INT-FWS)
March 14, 1963 and
Fed. Spec. PP-C-401a
April 25, 1952

FEDERAL SPECIFICATION

CLAMS, CHILLED, FRESH, AND FROZEN: SHUCKED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for the method of preparation and of packaging fresh, raw, shucked clams.

1.2 Classification. The clams shall be of the following types, classes, species, and sizes, as specified (see 6.1):

1.2.1

Types	Time from shucking to delivery
I — Fresh (chilled)	Less than 72 hours.
II — Frozen	Less than 180 days, as specified (see 6.1).

1.2.2 *Classes, species, and sizes.*

Class 1, *hard:*

(a) Hard, round, or quahog (*Venus mercenaria*, Atlantic) (*Mercenaria mercenaria mortoni*, Florida) for type II only.

Sizes of hard clams

Shucked clams per gallon (type I)	Shucked clams per 6-pound pack (type II)	Designation
Number	*Number*	
Up to 110	Up to 82	Chowders
110 to 175	82 - 131	Mediums
175 to 250	131 - 188	Cherry stones
Over 250	188 -	Little necks

Approximate weight for all sizes: In the shell, 80 pounds per bushel; shucked, 8 pounds per gallon.

(b) Ocean quahog (*Arctica islandica*, Atlantic).
Size: No size applicable.

(c) Butter (*Saxidomus nuttali*, Pacific).
Size: No size applicable.
Approximate weight: In the shell, 100 pounds per sack or 80 pounds per box.

(d) Little neck (*Protothaca staminea*, Pacific).
Size: No size applicable.
Approximate weight: In the shell, 100 pounds per sack or 80 pounds per box.

(e) Cockle (*Cardium corbis*, Pacific).
Size: No size applicable.

Class 2, *soft:*

(a) Soft-shell clam, sand clam, nannynose, or maninose (*Mya arenaria*, Atlantic and Pacific).

Sizes of soft clams

Clams shucked per gallon (type I)	Clams shucked per 6-pound pack (type II)	Designation
Number	*Number*	
Up to 800	Up to 225	Large
800 - 500	225 - 375	Medium
Over 500	375 -	Small

Approximate weight for all sizes: Shucked, 8 pounds per gallon.

Class 3, *surf or skimmer:*

(*Spisula solidissima*, Atlantic).
Size: No size applicable. Available minced.

2. APPLICABLE DOCUMENTS

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2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

PPP-B-636—Box, Fiberboard.
PPP-C-29—Canned Subsistence Items, Packaging and Packing of.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-668—Minimum Sanitary Standards for Food Establishments.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Governmental:

Federal Food, Drug, and Cosmetic Act and General Regulations for its Enforcement.
United States Public Health Service Publication No. 33, Cooperative Program for the Certification of Interstate Shellfish Shippers, Part I—Sanitation of Shellfish Growing Areas. Part II—Sanitation of the Harvesting and Processing of Shellfish.
United States Public Health Service Publication No. 956. Public Health Service Drinking Water Standards.

(Copies of the above three publications may be obtained upon application, accompanied by money order, coupon, or cash to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Prices may be obtained from the Superintendent of Documents.)

Nongovernmental:

Official Methods of Analysis of the Association of Official Agricultural Chemists.

(Copies are published by the Association of Official Analytical Chemists, Post Office Box 540, Benjamin Franklin Station, Washington, D. C., 20044.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Clams.* The clam meats shall be prepared from sound, strictly fresh clams which have been harvested from approved areas under the conditions set forth in Parts I and II of the Cooperative Program for the Certification of Interstate Shellfish Shippers of the United States Public Health Service Publication No. 33 where they are applicable.

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Dead clams (gapers) shall be discarded and shall not be shucked.

3.1.2 *Water.* The water used in rinsing and other processing operations shall conform to the U.S. Public Health Service Drinking Water Standards, Publication No. 956.

3.2 **Processing.** Clams shall be shucked as soon as possible after being landed. Shucking operations shall be separated from all further processing operations. The total time the clams are in contact with water shall not exceed 30 minutes. The time of blowing shall not exceed 5 minutes and shall count as at least 10 minutes in computing the total time of 30 minutes. The blowing of clams shall only be permitted at the shucking plant; soaking is prohibited. Meats shall be thoroughly drained, for a period of not less than 5 minutes, prior to being packaged. Shucked clams shall be packaged and chilled to an internal clam temperature of 40°F. or lower (not frozen) within 4 hours after shucking, and shall have an internal clam temperature of 32° to 38°F. within 6 hours after shucking. The shucked clams shall not be placed in direct contact with ice, except as provided by the U. S. Public Health Service Publication referenced in 3.1.1.

3.2.1 *Plant approval.* Raw clams shall be prepared only in plants certified by a State whose control measures have been endorsed by the U. S. Public Health Service and shall conform to all standards and provisions set forth in the Publication of the Public Health Service referenced in 3.1.1.

3.3 Finished product. The finished product shall consist of clean, wholesome, adequately drained (see 3.2) shucked (shells removed) clams of solid pack. The clams shall be free from adulteration, impurities, unapproved chemical preservatives (see 3.5), and added water. The shucked clams shall be free from presence of dark skin of the siphon greater than 1/2 inch in largest dimension, seaweed or algae, sand, and other foreign matter and shall contain not more than 4 pieces of shell per quart. Shucked clams shall be free from off odors such as evidenced by stale, sour, bilgy, petroleum, ammoniacal odors or flavors. Meat from clams shall be firm, not soft. Clam-meat color that is not considered characteristic of the species, such as gray, blue-black, or black, shall not be acceptable. There shall be no more than 5 cut, torn or broken clams per pint, and the broken clams shall not be included in determining the count for size. Type I (chilled) or type II (frozen) clams shall contain not more than 8 percent by weight of free clam liquor within 1 hour from time of packing (see 4.4.2). Free liquor for type II shall be tested prior to freezing (see 4.4.2).

3.3.1 *Definition of terms.*

3.3.1.1 *Gaper.* A dead clam in which the values are parted and will not close when the clam is disturbed.

3.3.1.2 *Cut or torn clam.* A clam with a cut, tear, or tag end greater than 1/4 the maximum length of the clam.

3.3.1.3 *Broken clam.* A clam that has a portion severed from the clam whose cut surface at the point of severence measures more than 1/2 the width of the clam.

3.4 Handling requirements.

3.4.1 *Type I, fresh (chilled).* Fresh clams shall be processed in compliance with 3.2 and 3.2.1 and shall be held and delivered to destination at a temperature of 32° to 38°F. Fresh chilled clams shall be delivered at destination not more than 72 hours after being shucked. Containers larger than 1 gallon (approximately 8 pounds) capacity shall not be used for type I clams. Type I clams shall have a minimum net weight of 8 pounds per gallon container.

3.4.2 *Type II, frozen.* Fresh clams shall be processed and packed in compliance with 3.2 and 3.3. The clams shall then be placed in a freezer equipped with forced air circulation, or equivalent, and frozen solid to an internal clam temperature of 0°F. or

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lower within 24 hours from the time of packaging. The clams shall be stored in such a manner that the internal clam temperature is not higher than 0°F. from the time the clams are frozen through the time of delivery. The clams shall be frozen not longer than 180 days before being delivered. The clams shall be of excellent condition and in a thoroughly frozen state and at time of delivery, shall exhibit no evidence of having been thawed and refrozen. Type II clams shall have a minimum net weight of 6 pounds per gallon container.

3.5 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act, and General Regulations for Its Enforcement.

3.6 **Sanitation.** The processing plant, grounds, equipment, personnel practices, and sanitary practices used in the production of this product shall be such as to minimize the possibility of contamination of the product through microbial growth, dust, pests, rodents, condensate, or other unsanitary sources. The product shall be prepared and processed with minimum delay between the various steps in production. (For military procurement only, the product shall be processed in establishments meeting the requirements of MIL-STD-668.)

4. QUALITY ASSURANCE PROVISIONS

4.1 **Preproduction sample approval.** When specified (see 6.1), a sample of finished product shall be submitted to the contracting officer for approval before production is commenced.

4.2 **Sampling.** Samples of finished products covered by this specification and samples of any materials, components, or constituents entering into the preparation of the finished products may be taken at any time by the Government inspector to determine compliance with the requirements.

4.3 **Inspection.**

4.3.1 *General.* Inspection shall be made by the procuring agency or a duly authorized representative at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after delivery to the point of destination.

4.3.2 *Responsibility for inspection.* The supplier is responsible for meeting all the provisions of the specification prior to submission to the Government for acceptance. Unless otherwise specified in the contract or order, the supplier is required to perform the examination and tests prescribed herein, using his own or any other inspection facilities and services acceptable to the Government.

4.3.2.1 Inspection records of the examination and tests shall be complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.3.3 *Preacceptance inspection (plant sanitation inspection).* Award of contracts shall be limited to plants which have received prior sanitary approval by the Military Veterinary Services or to plants operating under the continuous inspection supervision of the Bureau of Commercial Fisheries, Fish and Wildlife Service, United States Department of Interior, as officially listed by that agency.

4.3.4 *Sampling for inspection.* Sampling for inspection shall be in accordance with MIL-STD-105 except as otherwise indicated hereinafter.

4.3.5 *Component and material inspection.* In accordance with 4.3.2, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and stand-

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ards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3.6 *Process examination (for military procurement).* Process examination shall be performed to ascertain compliance with process requirements of 3.2 and 3.4 as concerns shucking operations, markings, blowing, draining, and refrigeration. Time of contact with water, blowing, draining and refrigeration records shall be maintained. Nonconformance to one or more of the above requirements reflected either by actual examination or by records shall be cause for rejection of the involved quantity of finished product. In addition, examination shall be performed during shucking operation to determine if gapers (see 3.3.1.1) are being used. The inclusion of gapers shall be cause for rejection of the involved quantity of product.

4.3.6.1 Inspection shall be made to assure that clams are obtained from approved areas as specified in 3.1.1. Nonconformance to the above requirement shall be cause for rejection of the involved quantity of finished product.

4.3.7 *Examination of end item.* Classification of defects found during examination shall be in accordance with table I through III, 4.3.7.1 and 4.3.7.2. The inspection levels, and the acceptable quality levels (AQL's), for the above tables shall be as shown below. The sample unit shall be 1 gallon of clams or one 6-pound can when type II is specified. Lot size shall be expressed in terms of number of gallons of type I product, or number of 6-pound cans if the product is type II. AQL's shall be expressed in defects per hundred units:

Table	Inspection level	AQL's	
		Major	*Minor*
I	S-3	2.5	10.0
II	S-3	—	10.0
III	S-3	2.5	10.0

TABLE I. *Examination for net weight and free liquor* [1]

Categories		Defects
Major	Minor	
	201	More than 1/2 pound under specified weight.[2]
151		More than 8 percent, by weight, of free liquor when drained for 2 minutes in accordance with 4.4.2.[3]

[1] When type II product is specified, examination shall be prior to freezing.

[2] Lot shall be rejected if sample data indicates a lot average net weight less than the specified net weight.

[3] The lot shall be rejected if sample data indicates lot average drained weight exceeds 8 percent.

TABLE II. *Examination for size (count) of clams per gallon* [1]

Category	Defects
Minor	
201	Number of clams not within range specified (applicable to all classes and species).

[1] Applicable to 6- and 8-pound containers (types I and II products).

TABLE III. *Examination of product characteristics* [1] [2]

Category		Defects
Major	*Minor*	
151		Grey, blue-black, or black clam meat.
152		More than 4 pieces of shell per quart.
153		Meat from clams not firm.
	201	Presence of seaweed.
	202	Presence of dark skin of the siphon greater than 1/2 inch, in largest dimension where applicable.
	203	More than 5 cut, torn or broken clams per pint.[3]

[1] Off condition or off odor such as pollution odor or odor of decomposition, petroleum odor, or ammoniacal odors, or product that is adulterated, that is, unapproved chemical preservatives or added water, shall be basis for rejection of the entire lot.

[2] Presence of foreign material, that is, dirt, grease, glass, metal, etc., shall be basis for rejection of the entire lot.

[3] See 3.3.

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4.3.7.1 *Examination of packaging, packing, labeling, and marking.* Examination of packaging, packing, labeling, and marking shall be in accordance with the examination criteria of PPP-C-29 or reference documents cited therein, as applicable.

4.3.7.2 *Examination of shipping containers.* Examination of the filled and closed shipping containers shall be in accordance with the examination criteria of PPP-B-636 for levels A and B pack. In addition, the following defects shall be incorporated in the table of examination: Major: marking, missing, incorrect, or illegible, Minor: arrangement not as specified. For level C pack, examination shall be in accordance with the examination criteria of PPP-C-29.

4.3.8 *Examination for temperature at time of shipment and delivery.* The product shall be examined for compliance with the provisions of 3.4.1 and 3.4.2, as applicable. Finding of 1 or more units of products in noncompliance shall be the cause for the rejection of the shipment.

4.3.9 *Final inspection.* Unless otherwise specified by the contracting officer, final inspection shall be made after delivery to the point of destination (see 6.5).

4.3.10 *Standby test samples.* The Government reserves the right to withdraw and hold standby test samples of components, or of the finished product, or of both (the quantity of which shall be not more than twice that required by this specification) for inspection purposes. These samples, if not used, shall be returned to the vendor.

4.4 Tests.

4.4.1 *Chemical analyses.* Unless otherwise specified (see 6.1), chemical analyses, if required by the purchaser, shall be made in accordance with the Official Methods of Analysis of the Association of Official Agricultural Chemists in effect on date of invitation for bids or in accordance with methods that give equivalent results.

4.4.2 *Test for free clam liquor.* Within 1 hour from the time the clams are packaged, they shall be tested for free liquor as follows:

a. Weigh the clams plus container.
b. Weigh the dry strainer or skimmer.
c. Spread the clams evenly over the strainer or skimmer (which has an area of not less than 300 square inches per gallon of clams and which has perforations of at least 1/4 inch in diameter and not more than 1-1/4 inches apart, or perforations of equivalent areas and distribution).
d. Allow the clams to drain for 2 minutes.
e. Dry the empty container and weigh it.
f. Weigh the drained clams and the strainer (or skimmer).
g. Calculate the percent free liquor as follows:

Weight of undrained clams = "U" = a-e
Weight of drained clams = "D" = f-b

$$\text{Percent of free liquor} = \frac{U - D}{U} \times 100$$

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be in accordance with 5.1.1 or 5.1.2, as specified (see 6.1).

5.1.1 *Level A.* Type II product shall be packaged in accordance with PPP-C-29, except that friction top reclosable cans may be used.

5.1.2 *Level C.* Unless otherwise specified in the contract or order, commercial packaging will be acceptable, except that for type II product only, hermetically sealed cans, as specified in 5.1.1, will be acceptable.

5.2 Packing. Packing shall be in accordance with 5.2.1 or 5.2.2, as specified (see 6.1).

5.2.1 *Level B.* Type II product shall be packed in accordance with PPP-C-29.

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5.2.2 *Level C.* The product shall be packed in containers to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation. Type I clams, packaged as specified in 5.1.2, shall be packed in ice in such a manner as to insure that containers are completely surrounded with ice

5.3 **Labeling and marking.**

5.3.1 *Civil agencies.*

5.3.1.1 *Unit containers.* Any commercial labeling or additional labeling, as specified, that complies with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder is acceptable.

5.3.1.2 *Marking of shipping containers.* In addition to markings specified in the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.*

5.3.2.1 *Unit containers.* Cans shall be labeled in accordance with PPP-C-29. In addition, all unit containers shall contain the following information:

Name of establishment or state certificate number.
Size A count.
Keep frozen or keep refrigerated (as applicable).

5.3.2.2 *Shipping containers.* In addition to the marking specified and any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129 for type I product and in accordance with PPP-C-29 for type II product. In addition, the following information, as applicable, shall be marked on the side and the top of the container in letters between 1 and 1-1/2 inches high (all letters being of the same height):

For chilled clams-(Keep refrigerated between 32° and 38°F.) *or* (Keep iced).
For frozen clams- Keep frozen 0°F.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, species, and size required and limited time of frozen storage (see 1.2.1 and 1.2.2).
(*c*) Whether preproduction sample is to be furnished (see 4.1).
(*d*) When chemical analyses are required (see 4.4.1).
(*e*) Levels of packaging and packing required (see 5.1 and 5.2).

6.2 **Bid samples.** If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample shall be definitely stated.

6.3 This specification does not include all types, classes, species, and sizes of clams that are commercially available, but is intended to cover the types, etc., which are suitable for Federal Government requirements (see 1.2.1 and 1.2.2).

6.4 **Contractor's inspection.** The contractor shall be required to assure that the product conforms with this specification prior to submission to the Government for final inspection.

6.5 **Destination inspection.** When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quality, and also to determine that on the basis of the code marks, the identical parcels previously examined and accepted have been delivered.

6.6 **Transportation descriptions.** Transportation descriptions and minimum weights applicable to this commodity are:

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Rail:

Clams, shucked, fresh or frozen.
Carload minimum weights 15,000 and 24,000 pounds.

Motor:

Clams, shucked, fresh or frozen.
Truckload minimum weight 15,000 and 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—GL
Navy—SA
Air Force—67

Review:

Army—MD
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Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

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